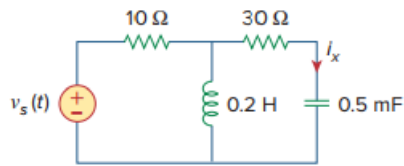


Problem

Given that $v_s(t) = 20 \sin(100t - 40^\circ)$ in Fig. 9.55, determine $i_x(t)$.

Figure 9.55:

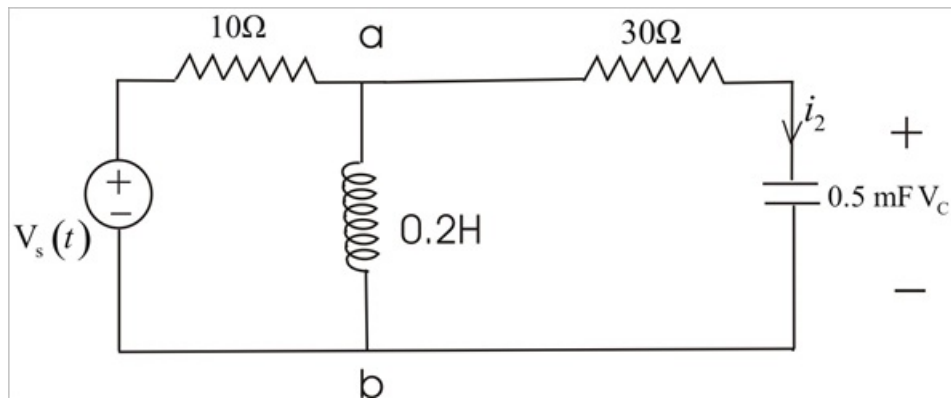


Step-by-step solution

Step 1 of 4

Given $V_s = 20 \sin(100t - 40^\circ)$

Find $i_x(t)$.



Step 2 of 4

$$\omega = 100$$

$$Z_1 = 30 + \frac{1}{j \times 100 \times 0.5 \times 10^{-3}} = 30 - j20$$

$$Z_2 = \frac{(30 - j20)(j \times 100 \times 0.2)}{30 - j20 + j20} = \frac{(30 - j20)(j20)}{30}$$

$$= \frac{(400 + 600j)}{30}$$

$$= 13.33 + 20j$$

$$Z_{eq} = 10 + 13.33 + 20j = 23.33 + 20j$$

$$V_s = 20 \sin(100t - 40^\circ)$$

$$= 20 \cos(100t - 40^\circ - 90^\circ)$$

$$= 20 \cos(100t - 130^\circ)$$

$$= 20 \angle -130^\circ$$

$$= I Z_{eq}$$

Step 3 of 4

$$\begin{aligned}
 I &= \frac{V_s}{Z_{eq}} = \frac{20 \angle -30^\circ}{30.72 \angle 40.60^\circ} \\
 &= 0.65 \angle -170.6^\circ \\
 V_{Z_2} &= I Z_2 \\
 V_{Z_2} &= I Z_2 = (0.65 \angle -170.6^\circ)(24.03 \angle 56.31^\circ) \\
 &= 15.619 \angle -113.69^\circ \\
 V_c &= \frac{1}{\frac{j \times 100 \times 0.5 \times 10^{-3}}{30 + \frac{1}{j \times 100 \times 0.5 \times 10^{-3}}}} \\
 &= \frac{-j20}{30 - j20} \{15.619 \angle -113.69^\circ\} \\
 &= \frac{-j20(30 + j20)}{1300} \{15.619 \angle -113.69^\circ\} \\
 &= (0.307 - j0.46) \{15.619 \angle -113.69^\circ\} \\
 &= (0.55 \angle -56.36^\circ)(15.619 \angle -113.69^\circ) \\
 &= 8.65 \angle -170.05^\circ
 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

angle should be -130 in the first line

Step 4 of 4

$$\begin{aligned}
 I_x &= \frac{V_c}{\frac{1}{5 \times 100 \times 0.5 \times 10^{-3}}} \\
 &= \frac{8.65 \angle -170.05^\circ}{-j20} \\
 &= \frac{8.65 \angle -170.05^\circ}{20 \angle -90^\circ} \\
 I_x &= 0.432 \angle -80.05^\circ
 \end{aligned}$$

In time domain,

$$i_x = 0.432 \cos(100t - 80.08^\circ)$$